

UTKARSH JOSHI

Nainital, Uttarakhand | +91 73027 28952 | joshiutkarshnt@gmail.com | linkedin.com/in/utkarsh-joshi-3037a6377 | github.com/UtkarshJoshiNt

OBJECTIVE

First-year CS student with a focus on systems programming, numerical computing, and software development. Started with Linux to solve a hardware problem in school, which led to C, C++, and OpenGL. Looking for internships in software engineering.

SKILLS

Languages: C, C++, Python, Bash

Systems & Tools: Linux/WSL2, Git, CMake, Docker, OpenMP

Frameworks & Libraries: FastAPI, CUDA, pybind11, NumPy, SciPy

PROJECTS

Astrosis - High-Performance Orbital Simulation Engine (C++17, CUDA, Python, pybind11)

- CLI + TUI + Python API for satellite orbit propagation, conjunction screening, and pass prediction; published on [PyPI](#) (pip install astrosis).
- RK4 integrator with J2-J4 perturbations, atmospheric drag, SRP, and lunisolar third-body forces; FP64 throughout for <0.1 km propagation error at 24h.
- Auto-selects fastest compute backend (CUDA, C++/OpenMP, NumPy, Python); benchmarked on RTX 2050 + Ryzen 5: C++ runs 5k-satellite batch in 55ms vs 36.8s in pure Python (676x).
- CUDA conjunction screening via 2-phase coalesced kernel: 400x400 object pair scan in 125ms vs 26.6s in Python (214x) on same hardware; C++ uses KDTree broadphase + Brent TCA refinement.
- Built on AutoCM, a prototype from NSH 2026 at IIT Delhi (Top 10 Finalist); expanded to support 10,000+ object simultaneous simulation.

Quip - Lightweight Unix Shell (C99, Linux)

- Unix shell written in C17 with zero external dependencies (libc only); ~39KB stripped binary, ~2ms startup, ~1.8MB RSS.
- Pipelines, I/O redirection, job control (fg/bg/ &), tab completion, and command history with circular-buffer disk persistence via fork/exec/waitpid.
- Python plugin system via Unix domain socket daemon; shell startup is unaffected and plugins are loaded on first use.

EDUCATION

Graphic Era Hill University, Bhimtal, Uttarakhand

B.Tech in Computer Science and Engineering | SGPA: 8.71 | *Expected May 2029*

ACHIEVEMENTS & ACTIVITIES

- Top 10 Finalist, National Space Hackathon (NSH) 2026, organized by IIT Delhi.
- Self-study: MIT Missing Semester of CS (completed); currently reading CS:APP (Bryant & O'Hallaron).